hormone (ICSH). In males, ICSH acts on the interstitial cells of testis to produce the male sex hormone, testosterone. In females, LH along with FSH matures the ovarian follicles. LH independently induces ovulation, maintains the corpus luteum and promotes synthesis and release of ovarian hormones. FSH and LH are collectively referred as gonadotropins. FSH and LH are not produced during childhood. The secretion of FSH and LH starts only during pre pubertal period.

vi) Luteotropic hormone (LTH): LTH is also called luteotropin or lactogenic hormone or prolactin or mammotropin. It is a protein hormone which stimulates milk secretion after the child birth in females. High prolactin secretion during lactation suppresses LH secretion and ovulation since it induces the corpus luteum hence named as luteo tropic hormone.

Hormones of neurohypophysis

i) Vasopressin or antidiuretic hormone (ADH) : ADH is a peptide hormone which promotes reabsorption of water and electrolytes by distal tubules of nephron and thereby reduces loss of water through urine. Hence it is called as anti diuretic hormone. It also causes constriction of blood vessels when released in large amount and increases blood pressure. ADH deficiency causes ***Diabetes insipidus*** which induces the production of large amount of urine.

ii) Oxytocin (means quick birth): It is a peptide hormone which stimulates vigorous contraction of the smooth muscles of uterus during child birth and ejection of milk from the mammary glands.

Vasopressin and oxytocin are composed of nine amino acids and are almost identical and they differ in only **two amino acids** and yet they have dramatically different physiological effects.

Amino acid sequence of **vasopressin**: cysteine-tyrosine-**phenyl alanine**-glutamine-aspargine-cysteine-proline-**arginine**-glycine.

Amino acid sequence of **oxytocin**:cysteine-tyrosine-**isoleu sine**-glutamine-aspargine-cysteine-proline-**leucine**-glycine.

Pituitary gland is located in a depression in the sphenoid bone of skull below the brain, so is also called hypothalamus cerebri. Discuss the following :

a. Pituitary gland is commonly called "master gland" of the body.
b. Discuss the role of hypothalamus and pituitary as a coordinated unit in maintaining physiological processes.
c. How does the posterior lobe of pituitary help in osmoregulation?

11.2.3 Pineal gland

In human, the pineal gland or epiphysis cerebri or conarium is located behind the third ventricle of brain and is formed of parenchymal cells and interstitial cells. It secretes the hormone, **melatonin** and its precursor serotonin. Melatonin induces sleep and serotonin induces waking up thus playing a central role in the regulation of circadian rhythm of our body.

It maintains the normal sleep wake cycle. It also regulates the timing of sexual maturation of gonads. In addition melatonin also influences metabolism, pigmentation, menstrual cycle and defence mechanism of our body.

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